Part-A Answers

1	2	3	4	5	6	7	8	9	10
a	d	b	c	c	a	b	a	d	b
11	12	13	14	15	16	17	18	19	20
a	b	a	d	c	b	a	d	c	c

Part – B Answer the following questions in one or two sentences:

21. If $62 = 34 + 4x$ what is x? (Answer :x is 7)

22. Given the demand function $q = 150 - 3p$, derive a function for MR.

23. Find the average cost function where $TC = 60 + 10x + 15x^2$

24. The demand function is given by $x = 20 - 2p - p^2$ where p and x are the price and the quantity respectively. Find the elasticity of demand for $p = 2.5$.

25. Suppose the price p and quantity q of a commodity are related by the equation $q = 30 - 4p - p^2$ find (i) e_d at $p = 2$ (ii) MR

26. What is the formula for elasticity of supply if you know the supply function?

27. What are the Main menus of MS Word?

Part – C Answer the following questions in one paragraph:

28. Illustrate the uses of Mathematical Methodsm in Economics.

29. Solve for x quantity demanded *if* $16x - 4 = 68 + 7x$. (Ans: x is 8)

30. A firm has the revenue function $R = 600q - 0.03q^2$ and the cost function is $C = 150q + 60{,}000$, where q is the number of units produced. Find AR, AC, MR and MC. (Answersa:AR = 600 – 0.03q ; MR = 600 – 0.06 q; AC = 150 + (60000/q))

31. Solve the following linear equations by using Cramer's rule.

$x_1 - x_2 + x_3 = 2$: $x_1 + x_2 - x_3 = 0$: $-x_1 - x_2 - x_3 = -6$

32. If a firm faces the total cost function $TC = 5 + x^2$ where x is output, what is TC when x is 10?

33. If $TC = 2.5q^3 - 13q^2 + 50q + 12$ derive the MC function and AC function.

34. What are the steps involved in executing a MS Excel Sheet?

Chapter-12.indd 272 02/01/2023 16:09:04